SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 3.3: Animal Gaseous Exchange and Respiration — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.3: Animal Gaseous Exchange and Respiration
Driving Question	DRIVING QUESTION: Why does my body work so much harder to get oxygen into my cells during exercise — and what happens inside those cells when there is not enough?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Your Body Changes When You Move — Anchoring the Phenomenon**	Learners measured and recorded their own resting breathing rate and pulse rate, then repeated measurements immediately after vigorous exercise (e.g., jumping jacks or running on the spot). Class data were pooled on the board, revealing a consistent, measurable increase in both rates across all learners. Teacher should highlight that this is real, personal evidence — not a textbook claim.	Both breathing rate and pulse rate increase measurably and universally during vigorous physical exercise. This is not random variation — it is a reliable physiological response. The class data set becomes the anchoring phenomenon for the entire sub-strand: something in the exercising body demands more oxygen and produces more waste gases, driving both the lungs and heart to work harder.	The increase in breathing and heart rate during exercise is the body's response to increased ATP demand in muscle cells. As muscles contract repeatedly, they consume oxygen and glucose faster and produce more carbon dioxide. Elevated CO_2 in the blood is detected by chemoreceptors, which signal the respiratory and cardiovascular centres in the medulla oblongata to increase rate and depth of breathing and cardiac output. Teachers should note: learners do not yet need to know the full mechanism — this lesson is purely about establishing the phenomenon and generating curiosity questions for the Driving Question Board.
Lesson 2 **Characteristics of Respiratory Surfaces — What Makes a Good Gas Exchange Surface?**	Learners examined diagrams, microscope images, or models of different respiratory surfaces (fish gills, insect tracheae, frog skin, human alveoli) and identified visible structural features. They compared surface area, wall thickness, moisture, and blood supply across the examples. A useful pedagogical move is to ask learners to rank the surfaces by efficiency before revealing	All efficient respiratory surfaces share four structural features: (1) large surface area — to maximise the number of diffusion sites; (2) thin walls, one to a few cells thick — to minimise the diffusion distance for gases; (3) a moist surface — gases must dissolve before diffusing across the membrane; (4) a rich blood supply (or equivalent fluid transport) — to maintain steep	These four features are adaptations that maximise the rate of diffusion, governed by Fick's Law: rate of diffusion is directly proportional to surface area and concentration gradient, and inversely proportional to diffusion distance. Every respiratory surface studied in this sub-strand — including human alveoli — can be evaluated against these four criteria.

	the four key criteria.	concentration gradients by continuously removing oxygen and delivering carbon dioxide.	Teachers should connect back to the driving question: during exercise, the steep concentration gradient needed for fast gas exchange is maintained because blood flow to the alveoli increases, pulling oxygen in and washing CO_2 away faster.
Lesson 3 **Structure and Adaptations of Respiratory Surfaces**	Learners studied labelled diagrams and/or prepared slides of the human alveolus, fish gill lamellae, insect tracheal system, and frog skin. They completed a comparison table, matching each organism's respiratory surface to the four criteria from Lesson 2. Observation of the counter-current flow in fish gills (if covered) is particularly striking — teachers should allow learners time to trace the direction of water and blood flow.	All efficient respiratory surfaces exhibit the same four key structural adaptations: large surface area, thin walls for a short diffusion pathway, moist surfaces, and a rich blood or fluid supply. In humans, the ~300 million alveoli provide a total surface area of approximately 70 m^2 — roughly the size of half a doubles tennis court. Fish gills use counter-current flow to maintain a concentration gradient along the entire gill surface, extracting up to 80% of dissolved oxygen from water.	Structure and function are directly linked: each adaptation of a respiratory surface can be explained by how it maximises one or more variables in Fick's Law. Teachers should prompt learners to apply this explicitly — e.g., 'Which adaptation reduces diffusion distance? Which adaptation maintains the concentration gradient?' Connecting to the driving question: when a Kenyan athlete like Faith Kipyegon sprints at high altitude in Eldoret, the reduced partial pressure of oxygen at altitude lowers the concentration gradient across her alveoli, making gas exchange less efficient — her body compensates over time by producing more red blood cells.
Lesson 4 **Mechanisms of Gaseous Exchange in Humans — External Respiration**	Learners observed a bell-jar diaphragm model (or video demonstration) showing how the diaphragm's contraction increases thoracic volume and decreases pressure, drawing air inward. They traced pressure changes during inhalation and exhalation, then identified which muscles are active in each phase. A common misconception to address: air is not actively 'sucked in' — it flows down a pressure gradient.	Breathing is a mechanical, pressure-driven process. During inhalation: the diaphragm contracts and flattens, external intercostal muscles contract raising the ribcage up and out — thoracic volume increases, pressure drops below atmospheric, and air flows in. During exhalation (at rest): the diaphragm relaxes and domes upward, internal intercostal muscles (and abdominal muscles during forced exhalation) contract — thoracic volume decreases, pressure rises above atmospheric, and air flows out.	This is Boyle's Law applied to biology: pressure and volume are inversely related in a closed system. The lungs cannot move themselves — they are moved by the thoracic cavity. Teachers should use the Kenyan context of a Maasai warrior running long distances: deep, forceful breathing during a sprint recruits accessory muscles (sternocleidomastoid, scalenes) to achieve even greater thoracic expansion. Connecting to the driving question: the increased depth and rate of breathing during exercise directly increases the volume of fresh, oxygen-rich air reaching the alveoli per minute (minute ventilation = tidal volume × breathing rate).
Lesson 5 **Mechanisms of Gaseous Exchange in Humans — Alveolar & Cellular Exchange**	Learners traced the pathway of one oxygen molecule from inhaled air through the alveolus wall, into a red blood cell, through the circulatory system, out into a muscle cell, and into a mitochondrion. They also traced the reverse journey of CO_2. Diagrams showing partial pressure	Oxygen does not travel directly from the lungs to the muscles — it is carried by haemoglobin in red blood cells through the circulatory system. At the alveolus (external respiration): O_2 diffuses from high partial pressure in the alveolar air into the blood; CO_2 diffuses from high partial pressure in	Gas exchange at both sites is driven entirely by diffusion down partial pressure gradients — no energy is expended. The haemoglobin molecule loads O_2 at the alveolus (where pO_2 is high) and unloads it at the tissues (where pO_2 is low due to cellular respiration consuming it). Teachers

	gradients at the alveolus and at the tissue are central — learners should annotate these with arrows and partial pressure values (O_2: alveolar ~13.3 kPa, blood entering ~5.3 kPa; CO_2: blood entering ~6.1 kPa, alveolar ~5.3 kPa).	the blood into the alveolar air. At the tissues (internal respiration / cellular exchange): O_2 diffuses from the blood into respiring cells; CO_2 diffuses from cells (where it is produced) into the blood.	should note: during exercise, muscle cells consume O_2 faster, so pO_2 in the tissues falls further, steepening the gradient and accelerating unloading. This directly answers the first half of the driving question: the body works harder during exercise because steeper gradients and higher blood flow are needed to meet the increased O_2 demand of contracting muscle cells.
Lesson 6 **Aerobic Respiration: Unlocking Energy from Ugali Inside Your Cells**	Learners used the limewater CO_2 test on exhaled breath before and after exercise to confirm increased CO_2 production during exercise. They also observed (or were shown) the mitochondrial structure and connected it to the site of aerobic respiration. The Kenyan food context (ugali → maize starch → glucose) should be made explicit: glucose from the breakdown of ugali is the substrate entering cellular respiration.	Aerobic respiration occurs in the mitochondria of cells and uses glucose (from digested food such as ugali) and oxygen (delivered by the circulatory system) to release energy in the form of ATP. The overall word equation is: Glucose + Oxygen → Carbon Dioxide + Water + Energy (ATP). The balanced chemical equation is: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + ATP. It is a continuous, multi-step process (glycolysis → Krebs cycle → oxidative phosphorylation) that releases energy in controlled, usable steps.	Aerobic respiration is the primary reason the body needs a constant supply of oxygen: without it, the mitochondria cannot complete the electron transport chain and ATP production drops dramatically. The CO_2 produced is a direct metabolic waste product that must be expelled — this is why the limewater turned milkier faster after exercise. Teachers should connect this to the driving question: the harder the muscles work, the more glucose and oxygen they consume and the more CO_2 they produce — which is the direct chemical signal that drives increased breathing rate. The mitochondrion is therefore the central organelle of this entire sub-strand.
Lesson 7 **Anaerobic Respiration: Yeast Balloons, Burning Legs, and Oxygen Debt**	Learners set up yeast + glucose + warm water in a bottle with a balloon over the neck and observed balloon inflation over 15–20 minutes, confirming CO_2 production without oxygen. They also related the burning sensation in leg muscles during intense sprinting (e.g., during the 100 m at a school athletics day in Nairobi) to lactic acid accumulation. The two contexts — yeast fermentation and animal anaerobic respiration — should be kept clearly separate.	Anaerobic respiration in yeast (fermentation): Glucose → Ethanol + Carbon Dioxide + (small amount of ATP). Anaerobic respiration in animals (including humans): Glucose → Lactic Acid + (small amount of ATP). In both cases, far less ATP is produced than in aerobic respiration (net 2 ATP vs. ~36–38 ATP). In animal muscle cells, lactic acid accumulates during intense exercise when oxygen delivery cannot meet demand, causing muscle fatigue and the burning sensation. After exercise, extra oxygen (oxygen debt) is consumed to oxidise lactic acid back to glucose in the liver (Cori cycle).	Anaerobic respiration is a short-term emergency mechanism — it allows muscles to keep contracting when the cardiovascular system cannot deliver oxygen fast enough. The oxygen debt (excess post-exercise oxygen consumption, EPOC) explains why breathing remains elevated even after exercise stops: the body is still working to clear lactic acid and replenish ATP and phosphocreatine stores. Teachers should connect this directly to the second half of the driving question: 'What happens inside cells when there is not enough oxygen?' — the answer is anaerobic respiration, lactic acid build-up, rapid fatigue, and the oxygen debt that follows.
Lesson 8 **Respiratory Quotient (RQ): What Fuel Is Your Body Burning?**	Learners calculated RQ values from given or measured volumes of CO_2 produced and O_2 consumed for different fuel types. They compared theoretical RQ values:	The Respiratory Quotient (RQ) = Volume of CO_2 produced ÷ Volume of O_2 consumed. RQ values indicate which fuel substrate is being oxidised: RQ = 1.0 → carbohydrate;	RQ is a practical diagnostic tool: sports scientists and nutritionists use it to determine an athlete's metabolic state and fuel use. Teachers should note that RQ

	carbohydrate (RQ = 1.0), fat (RQ ≈ 0.7), protein (RQ ≈ 0.9). Kenyan food examples make the context concrete: maize starch from ugali (carbohydrate), fat from nyama choma (fat), beans/githeri (protein). A data-analysis exercise using a respirometer or provided data sets reinforces the calculation.	RQ ≈ 0.7 → fat; RQ ≈ 0.9 → protein. A mixed diet produces an RQ of approximately 0.85. Fats require proportionally more oxygen to oxidise (more C–H bonds), hence lower RQ. During intense exercise, RQ rises toward 1.0 as the body preferentially burns glucose for fast ATP production. An RQ above 1.0 indicates anaerobic respiration is occurring (excess CO_2 from buffering lactic acid).	above 1.0 is a detectable signal of anaerobic respiration onset — connecting back to Lesson 7. The Kenyan farming context is useful here: a subsistence farmer in the Rift Valley who eats primarily ugali (carbohydrate) will have an RQ close to 1.0 during moderate work, while a diet rich in nyama choma (fat) would yield a lower resting RQ. This lesson also reinforces the mathematical, data-analysis dimension of the NGSS practice of 'Analysing and Interpreting Data.'
Lesson 9 **Practical & Data Analysis: Factors Affecting the Rate of Respiration**	Learners used a simple respirometer (or yeast + glucose + coloured water manometer setup) to measure CO_2/O_2 consumption at different temperatures (e.g., 10 °C, 25 °C, 37 °C, 45 °C). They plotted results and identified the optimum temperature. The enzyme-temperature relationship and enzyme denaturation above the optimum should be explicitly observed in the data — rate increases then drops sharply. Results should be shared and class averages calculated to reduce experimental error.	The rate of cellular respiration is controlled by enzymes. Enzyme activity (and therefore respiration rate) increases with temperature up to an optimum of approximately 37 °C for human enzymes (yeast optimum is slightly lower, ~30–35 °C). Above the optimum, enzymes denature — the active site changes shape, substrate cannot bind, and respiration rate falls sharply. Below the optimum, kinetic energy is insufficient for frequent enzyme–substrate collisions, so rate is low. This is the same principle as any enzyme-controlled reaction studied in biochemistry.	Temperature is one of three major factors affecting respiration rate covered in Lessons 9–10. The enzyme-denaturation story is crucial: it explains why fever (high body temperature) is dangerous — it risks denaturing respiratory enzymes in cells. It also explains why cold-blooded animals (e.g., the Nile monitor lizard at Lake Victoria) become sluggish in cool water — their cellular respiration slows. Teachers should connect to the driving question: during intense exercise, muscle temperature rises, initially increasing enzyme efficiency and respiration rate — but if core temperature rises too high (heat stroke), enzyme denaturation becomes a life-threatening risk.
Lesson 10 **Factors Affecting Rate of Respiration (continued) & Importance of Gaseous Exchange**	Learners examined data sets or conducted brief experiments showing the effect of substrate concentration (glucose concentration) and oxygen availability on respiration rate in yeast or germinating seeds. They also reviewed evidence linking efficient gaseous exchange to survival and athletic performance — e.g., comparing oxygen consumption data for a resting person vs. a Kenyan marathon runner vs. a patient with pneumonia.	Three additional factors affect the rate of cellular respiration: (1) Substrate concentration — increasing glucose increases respiration rate up to the point where enzymes are saturated; (2) Oxygen availability — higher O_2 concentration increases aerobic respiration rate; in low O_2, cells switch to anaerobic respiration (less efficient); (3) The importance of gaseous exchange — efficient lung function, rich blood supply, and healthy respiratory surfaces are essential for delivering sufficient O_2 and removing CO_2. Diseases that damage respiratory surfaces (e.g., pneumonia, tuberculosis — TB is still prevalent in parts of Kenya) reduce gas exchange efficiency and limit aerobic	This lesson consolidates the link between the macroscopic (breathing, circulation) and microscopic (cellular respiration, enzyme kinetics) levels. Teachers should emphasise that all the factors affecting respiration rate ultimately work through enzyme kinetics — more substrate means more collisions, more O_2 means the electron transport chain can run fully, etc. The public health connection is powerful: TB destroys alveolar tissue, reducing surface area and increasing diffusion distance — both reducing gas exchange efficiency. Connecting to the driving question: a person with TB cannot meet the O_2 demands of vigorous exercise because their respiratory surfaces are compromised,

		respiration.	forcing earlier onset of anaerobic respiration.
Lesson 11 **Lesson 11**	Lesson 11 content was not provided in the source materials. Teachers should use this lesson to consolidate Lessons 6–10 through a structured review activity — for example, learners reconstruct the complete pathway from breathing in air at the alveolus to ATP production in the mitochondrion, annotating each step with the relevant factor that could limit it. A peer-teaching or gallery walk format is recommended.	Consolidation lesson: learners should be able to state the word and chemical equations for both aerobic and anaerobic respiration, define and calculate RQ, list and explain all factors affecting respiration rate (temperature, substrate concentration, oxygen availability), and describe the consequences of insufficient gaseous exchange. This lesson serves as a formative checkpoint before the summative review in Lesson 12.	Teachers should use this lesson diagnostically — identify and address remaining misconceptions before Lesson 12. Common misconceptions to probe: (1) Confusing breathing (ventilation) with cellular respiration; (2) Thinking anaerobic respiration produces no ATP; (3) Believing RQ is always fixed regardless of diet or exercise intensity. Each misconception should be addressed explicitly. Connect back to the driving question: by the end of this lesson, learners should be able to give a partial but substantive answer — they understand what happens at the cellular level; Lesson 12 integrates the full explanatory chain.
Lesson 12 **Summative Review & Transfer: Explaining the Body's Response to Exercise**	Learners revisited the original class data from Lesson 1 (breathing rate and pulse rate before and after exercise) and were asked to now explain every observation using the knowledge built across all 11 preceding lessons. They also applied their understanding to a novel transfer scenario — e.g., explaining why a footballer playing at high altitude in Eldoret tires faster, or why a patient recovering from pneumonia cannot climb stairs without becoming breathless. DQB questions from throughout the unit were reviewed and answered.	Learners integrated the complete explanatory chain: increased muscle ATP demand during exercise → increased rate of aerobic cellular respiration in mitochondria → increased O_2 consumption and CO_2 production → rising blood CO_2 detected by chemoreceptors → signals to increase breathing rate and depth (greater ventilation) and heart rate (greater blood delivery) → steeper partial pressure gradients accelerate alveolar and cellular gas exchange → if O_2 still insufficient, anaerobic respiration begins → lactic acid accumulates → muscle fatigue → post-exercise oxygen debt is repaid by continued elevated breathing. RQ rises toward and above 1.0 during intense exercise.	This lesson is the capstone of the sub-strand. Every concept — respiratory surface adaptations, ventilation mechanics, gas exchange by diffusion, aerobic and anaerobic respiration, RQ, factors affecting respiration rate — converges here to answer the driving question completely and precisely. Teachers should insist that learners articulate the full causal chain, not just isolated facts. The transfer tasks are essential: competency is demonstrated when learners can apply the explanatory framework to a new scenario (altitude, disease, different sports) rather than simply recalling facts. Celebrate the return to the Lesson 1 phenomenon — the class's own body data — as the most powerful pedagogical close to the unit.